



Automating the Generation of 3D Multiple Pipe Layout Design Using BIM and Heuristic Search Methods

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Abstract. The layout design and spatial coordination of multiple pipe systems is major challenge in the construction industry. The purpose of multiple pipe layout design is to find out an optimal layout for numerous individual pipes to route from a different start locations to different end locations in a 3D environment with no clashes under various kinds of constraints. Currently, pipe layout design is conducted manually by consultants, which is tedious, labor intensive, error-prone, and time-consuming. This paper proposes a BIM-based approach for layout design of multiple pipes using heuristic search methods. Algorithms are developed based on a directed weighted graph according to the physical, design, economical and installation requirements of pipe layout design. Clashes between pipes and with building components are considered and subsequently avoided in the layout optimization. Based on the developed algorithms, simulated annealing (SA) algorithm is used to approximate global optimization in a large search space for multiple pipe layout optimization. As for layout design, Dijkstra algorithm and two heuristic algorithms namely 3D A* and fruit fly optimization algorithm (FOA) are implemented and compared to obtain the multiple pipe system layout design. An example of a typical plant room with nine pipe routes is used to illustrate the developed approach on multiple pipe layout design. The result shows that the developed approach can generate optimal and clash-free multiple pipe system layout. Compared with the conventional method, the developed approach significantly reduces the time and cost for designing multiple pipe layout.

Keywords: Multiple pipe layout · Automatic design optimization · Building information modeling · Heuristic method

1 Introduction

Multiple pipe system layout design and spatial coordination plays an important and challenging role in construction industry. Efficient layout design of the pipe systems becomes more and more complicated due to numerous pipe systems involved in the layout space in compliance with various constraints such as physical, design, operational, and economical constraints [1]. The purpose of multiple pipe layout design is to find the most economical spatial arrangement of equipments and vessels and the route for interlinked pipes and accessories, as shown in Fig. 1, satisfying many constraints

such as physical, construction, operation, maintenance, safety and other various code requirements [2]. The design of pipe system layout needs to consider factors such as building space geometry, system requirements, design code specifications, and the locations and configurations of relevant equipments [3].

Pipe systems aims to transfer fluids from a starting point to a destination point with a certain time interval through an appropriate route. Pipe layout is very popular and play an essential part in our daily lives, for example, in electric power plants, chemical plants, factories, buildings, sewage, automobiles, etc. [4], and its contribution to the overall capital cost is often nontrivial. Pipe layout design is one the most time consuming, expensive and complicated steps in any pipe system design since it can take over 50% of the total design man-hours and all other detail design depends on it [5]. One of the most exhausting aspect of pipe layout design is to conduct pipe route planning to connect start and end points. Especially, when large pipes network layout is involved in the 3D space. Pipe routing is usually determined after the positions of the major components are determined which is done by as per some industry guidelines, building requirements, and designer and consultant experience [1]. As detail functional design is characteristically less creative and more routine than the earlier route planning and designing steps [6], a complete automation and design optimization of multiple pipe route layout might offer an attractive way to cut down on tedious and irksome work, leading to saving in time and money. Effective layout design of the pipe system is constrained by various consideration, such as [4]:

- Physical constraints: Avoidance of clash with structural elements, architectural elements, and other obstacle around; clearance for entry and exit requirements.
- Design constraints: clashes among pipes; branching of pipes, minimum equipment clearance requirements; buffer for danger zones; accessibility requirement for valves and accessories.

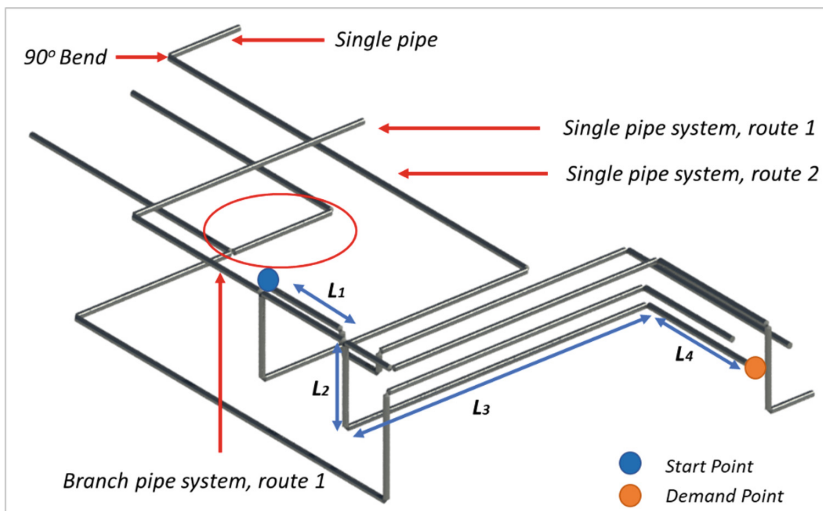


Fig. 1. Multiple pipe system layout

- Economical constraints: minimum pipe length and bends; installation cost (near to walls and obstacles for support); support sharing among pipelines.

Conventionally, engineers design the pipe route manually [7] or use physical model in which route planners interact with physical wires to find optimal route [8]. This conventional method requires huge time and cost with low accuracy. The use of computer-aided design (CAD) software has partially automated the manual process. Yet it requires human effort and decision making to plan, design and resolve the various design constraints. Also, different design aspects typically require different software's and the process is usually not well-integrated and hence making the design process expensive, tedious, labor intensive, error-prone, time-consuming and unadaptable to changes [9]. Moreover, the routing of pipes neither consider various pipe layout constraints nor considers all the possible solutions for the optimal layout that satisfy design requirements, thereby leading to uneconomical design. Physical pipe layout design is a complex spatial geometry problem. Since several solutions are possible for pipe routing design and checking each solution will be NP-Hard. Hence, a heuristic optimal layout technique for pipe system layout design is desired. Furthermore, the layout design gets more complicated and time consuming when the 3D space consists of multiple pipe systems. Therefore, a complete automation with heuristic design optimization approach is required to quickly and optimally plan multiple pipe layout in 3D environment.

With the advent of BIM technologies in the AEC industry, the conventional layout design process has been changed and improved with various potential design constraints addressed. BIM is used in this study because BIM can allow smooth integration of piping systems to architectural and structural envelope [10]. BIM employs digital representation of 3D parametric modeling of building to support the intelligent exchange of building information [11]. Furthermore, 3D BIM enables the designer to visualize and extract necessary geometry and semantic information of the space [12], location of equipments, start and end point of pipe system, and location of building (architectural and structural) components. Additionally, it can conceptualize routing of different pipe routes thus enabling quick identification of clashes and interference thus avoiding actual reworks on site. Besides, BIM provides a systematic approach to manage and support the intelligent exchange of building data information in digital format through the lifecycle of a building project [10]. Although BIM supports pipe design process, but still it lacks in clash avoidance at the stage of pipe routing, thus clash detection is required after design process followed by further modification, if clash exists, which is time consuming and tedious. Furthermore, pipe route as suggested by BIM application cannot guarantee selection of optimal path and still requires human intervention and consultant decision in choosing an appropriate pipe system layout. Therefore, this paper aims to develop an automated building information modeling (BIM) based heuristic approach for multiple pipe layout design optimization in 3D environment.

2 Literature Review

2.1 Multiple Pipe System Layout

Various sorts of automated strategies are developed and are widely used in present industrial practice, relatively less has been evolved on the layout routing problem in the literature. Several algorithms particle swarm optimization (PSO) [13], ant colony optimization (ACO) for ship building [14] and aircraft engines [3], genetic algorithm in plant layout (GA) [4], octree modeling and modified max-min ant system optimization algorithm in aero-engines [15] have been used for pipe routes planning. Yet, limited development has been made to pipe routing in built environment, as the constraints and requirements are distinct such as, objectives are of substantially different in nature, the constraints are often complex, and the routing is 3D. The researches mainly focus on single pipelines with two terminals with various imperceptive assumptions while developing algorithms. In addition, the optimization strategy only focused on clash-free pipe routes neglecting other factors such as economical, construction, design, operation, safety, etc. Siu and Niu [16] presented branch-pipe-routing approach for ships using improved genetic algorithm. However, the solutions obtained were only based on limited strategies thus could not be guaranteed as the optimal solution in terms industry practical constraints.

Although significant research efforts have been carried out to enhance the efficiency of pipe route planning in single or multiplanar, with fewer constraints, limited research have been focused on multiple pipe layout design in 3D. Also, the current researches do not focus on developing an automatic comprehensive strategy of extracting the details and information required to generate pipe layout planning, thus the process is still disintegrated and manual. Thereby leading to the process error prone due to lack of information flow. Therefore, this paper presents an integrated automated approach to generate optimal multiple pipe system layout in 3D environment.

2.2 Optimization Algorithm for Route Planning

Several algorithms have been developed to tackle NP-Hard optimization problems particularly when the environment is more complex and have shown better results in overcoming the limitation of classic methods [17]. Heuristic algorithms are most often employed when approximate solutions are enough and exact solutions are necessarily computationally expensive [18]. Global optimization algorithms have a goal of finding global optimum solutions. This generally involves more rigorous search than local optimization. However, the quality of the solutions obtained from the heuristic algorithms can be improved by spending additional time in the optimization process.

Some commonly used algorithms are Dijkstra's algorithm [19–21], Ant Colony Optimization has been used in path planning problems [22, 23], layout arrangement in marine engine room [24], pipe route design in ship design [14], aircraft engines [3], robot path planning [17]; Particle Swarm Optimization, it is a stochastic population-based optimization method, applied to solve many layout and routing problems such as cable routing optimization for offshore wind power plants [25], energy-competent routing protocol for MANETs [26], layout optimization for aero-engine [27], evacuation routing

optimization [28], pipe routing [13, 29], layout arrangement in marine engine room [24]; Fruit Fly Algorithm is one of the recently proposed swarm intelligence optimization algorithm for solving shortest path problems for Mobile Ad-Hoc Network [30], travelling salesman problem [31, 32]; Genetic Algorithm has been used in various fields for shortest path and planning problems such as path planning for mobile robot planning [33], pipe routing [3], route planning [34–36], ship design [16], mobile ad-hoc networks [37]; A* Algorithm [38, 39], and Simulated Annealing (SA) for path optimization [40, 41]. SA is better than other techniques in finding global optima as it often accepts and check worst solution, thereby allowing more extensive search for the global optimal solutions [42].

Many of these heuristic methods are being widely used for route and path planning with various limitation such as requires more computational time, resulting in unavailable path, local optimized route, less accurate results, demands additional modification, requires lot of memory space. However, the combination of these heuristic algorithms can be effectively used to overcome their inherent limitation and can be efficiently used to plan out global optimized routes in a complex environment. Therefore, this paper presents an approach for multiple pipe layout design using combination of algorithms. Simulated annealing (SA) algorithm is used to approximate global optimization in a large search space for multiple pipe layout optimization. As for layout design, Dijkstra algorithm and two heuristic algorithms namely 3D A* and fruit fly optimization algorithm (FOA) are implemented and compared to obtain the multiple pipe system layout design. Dijkstra is used for comparison in this paper because of its ability to find the optimum route. Though, the time required to find the optimum path by Dijkstra becomes remarkably long when the search scope is broad, still it can provide better foundation for comparison with heuristic algorithms.

3 Proposed Approach

The objective of multiple pipe layout design approach is to find out an optimal clash-free layout of different pipe routes (paths) from their respective start location to an end location in a 3D environment with interference under various kinds of constraints. The approach includes large number (numerous) of pipe systems in a 3D layout space having single pipelines with two terminals and branch pipelines with multiple terminal. The proposed approach consists of four modules (1) BIM modeling, (2) Space and constraints modeling, (3) Fitness function, and (4) Optimization Algorithms.

3.1 BIM Modeling

To design a pipe system layout, geometric and semantic information about building layout space is needed and BIM technology allows smooth flow of these information such as, (1) information related to layout space are required to understand the confined working space for pipe system layout, (2) building components or objects information are needed to avoid interference of pipe routes with various obstacles and thereby, generating clash-free pipe layouts, (3) location of entrances, exits and gateways in order to provide accessible passage for user, (4) position and dimension of equipments to avoid interference with them and to provide necessary buffer for laying pipe routes

around various equipments, (5) location coordinates of all the supply (start) and design (end) points as per the designer experience, (6) Selection of branch points as per the designer requirements. Branch points refers to the points adopting subdivision of pipe routes connecting one supply to many demands' points and many supplies to one demand point.

BIM is used to automatically extract and provide geometric and semantic information of 3D space for clash detection, spatial coordination, accessible routes and clearance, and multiple pipe system layout. In order to eliminate tedious human effort and errors, as well as to improve the efficiency of information flow, plug-ins for the BIM-platforms can be developed to automate the geometric and semantic information extraction. The information from BIM-model can be easily and automatically extracted and passed to an external file (e.g. spreadsheet, text document) for further design layout process.

3.2 Space and Constraint Modeling

Space modeling is a conceptual expression of space environment information. It is the key to obtain geometric information about the space environment quickly and accurately, which is important to generate a valid navigation graph. The graph, $G = (N, E)$ is represented as combination of nodes (N) and edges (E) where, each node represents possible junction or joints 3D points and edge represents the segment running between two nodes. This method reduces our search space to a simple 3D array. Each node and edge in the graph are composed of weights which corresponds to a cost value reflecting any limitation as per the constraint modeling on total piping layout. In general, the cost function of total pipe layout is represented by cost of edge and nodes inhabited by the individual pipe routes. Thus, selecting the edges and nodes with lowest cost from the start node to end node will lead to generation of pipe layout with minimum cost.

Constraint modeling refers to the consideration and elimination of various constraints such as physical, design, operational and economical constraints as mentioned in Sect. 1. According to economical constraints, the pipe system layout should have minimum length, less bends, routed around the building elements for necessary supports and close enough to share common pipe supports, racks and hangers during installation. The economical constraints can be tackled by the formation of appropriate optimization function as mentioned in next section.

To eliminate physical constraints i.e. clashes with the building (architectural and structural) elements and other equipments and objects within the defined building space are examined. The nodes which intersects with various physical components are restricted for the pipe routes and are regarded as unwalkable for pipe layout. For example, in Fig. 2 the node_{i+1} is restricted, as it clashes with the structural element and other pipe. In order to provide mandatory clearance for operational constraints, a buffer value based on piping code and requirements is selected and any nodes within that space is impeded and is restricted for routing of pipe layout. The clashes among various multiple pipe system, as shown in Fig. 2, are also tackled by a rule-based strategy and are discarded for pipe layout generation. Any clash of nodes and edges with already routed pipe systems during the layout generation is restricted as per the order of pipe system preference. For the branch pipe systems having common supply or end points

can have a common route up to an appropriate location (node) depending upon the other constraints requirements.

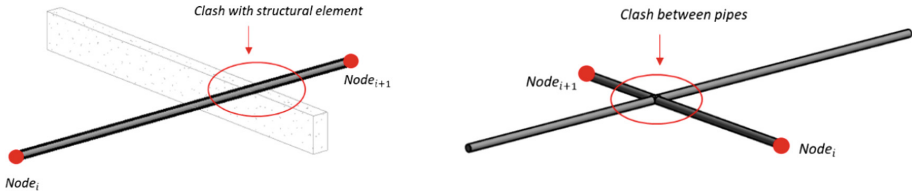


Fig. 2. Routing restriction due to clashes

3.3 Fitness Function

To evaluate the multiple pipe system layout for the optimal result, the total cost of the layout system is calculated in order to reduce its pressure loss by minimizing its length and bends, and installation cost, which is one of the major cost ignored while designing thereby leading to addition cost during installation or rework at the latter stage. The formulation of cost function for multiple pipe system layout is based on minimizing variables affecting pressure loss and installation cost.

Cost Due to Pressure Loss: The pressure loss suffered in the pipe system is caused due to the friction along the pipe length (straight path) and momentum exchanges results from the change in direction of the flow (bend) [44]. The friction along the pipe length is calculated as per the total distance travelled by the edges from the supply point to the demand point, following the pipe route. The distance travelled between the two nodes is calculated using Manhattan method [45], as shown in Eq. (1). The momentum exchanges are represented by the cost of nodes on which a bend is introduced. The cost of bend is calculated by the equivalent length method depending on the bend type. Equivalent method calculates the pressure loss through a bend as a length of straight pipe under the corresponding flow rate [44].

$$L_i = \text{abs}(X_i - X_{i+1}) + \text{abs}(Y_i - Y_{i+1}) + \text{abs}(Z_i - Z_{i+1}) \quad (1)$$

The cost function (C_p) based on pressure loss is calculated by the cost for movement for the start to end node (cost of edges and nodes), following the route, and is given as in Eq. (2):

$$C_p = C_m \left[\sum_{i=1}^n L_i + \sum_{j=1}^x NB_j \right] \quad (2)$$

where, C_p = Cost for movement from the start to end node (US\$); C_m = material cost of pipe per unit length (US\$/m); L_i = total distance travelled among the nodes (m); NB_j = equivalent length for a bend (j) (m); n = total number of nodes; x = total number of bends. In this study, 90° curved elbow is considered and thus, $NB_j = 30$.

Installation Cost: The installation cost refers the cost incurred in fitting pipe system on site and depend on the location of pipe system routes. Nearer (with minimum clearance) the pipe system with a supporting component (architectural and structural elements) to hold the pipe system, lesser is the installation cost. Hence, the total installation cost of a pipe system is evaluated as per the perpendicular distance of pipe route nodes from the nearest supporting elements such as wall, column, beam etc., as shown in Fig. 3 and is given by Eq. (3):

$$C_I = I_C \sum_{i=1}^n d_i \quad (3)$$

where, C_I = Installation cost of a pipe system; I_C = Installation Cost of pipe/unit length (US\$/m); d_i = perpendicular distance of node from the nearest supporting elements (i) from supporting elements, (m); n = total number of nodes.

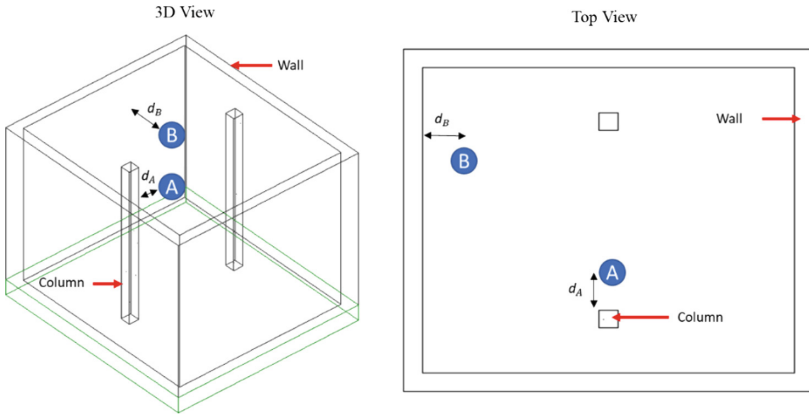


Fig. 3. Perpendicular distance form nearest supporting elements

As mentioned above the total cost of multiple pipe system layout (TCS) is evaluated by the summation of Cost due to pressure loss (C_P) and installation cost of a pipe system from start to end (C_I) and is given in Eq. (4). The result with minimum total cost is adopted to be the optimal multiple pipe system layout.

$$\text{Min } TCM = C_P + C_I = C_m \left[\sum_{i=1}^n L_i + \sum_{i=1}^n NB_j \right] + I_C \sum_{i=1}^n d_i \quad (4)$$

3.4 Optimization Algorithms

This paper presents two sets of optimization approaches one for individual pipe route optimization and other for finding global optimum layout among large search spaces.

Pipe Route Optimization aims to find optimal individual pipe route connecting a single supply and demand point. The approach considers two heuristic algorithm, namely A* and Fruit fly algorithm, and Dijkstra algorithm for pipe route optimization.

A* algorithm is a graph traversal greedy best-first-search algorithm and finds a least-cost path among possible solutions from start to end node using heuristic function [46]. As this paper demonstrate pipe system route in a 3D environment, the A* algorithm is modified to 3D A*. For the successful application of 3D A* algorithm, the 3D space is converted to weighted graph (nodes and edges). The nodes which are available for path is marked as walkable and the nodes which are not available for the path is marked as unwalkable.

As this study considers pipe route in the straight and orthogonal (90°) direction along the coordinated axis. During each search step in the 3D environment, there are six possible movement direction 2 in X-axis, 2 in Y-axis, and 2 in Z-axis. Path finding begin with the start node and moves to the other walkable neighbor nodes next to it until the end node is reached. The cost function of path as per A* algorithm fitness function as given in Eq. (4) and heuristic cost function. In this study, the Manhattan method [30] is used to determine the heuristic cost function shown in Eq. (5), as it is the standard heuristic method of square graph. The algorithm so far gives us shortest route based on the length (straight path). In this study, the 3D A* is modified and improved based on length, bends and the installation requirements.

$$h(n) = \text{abs}(X_c - X_t) + \text{abs}(Y_c - Y_t) + \text{abs}(Z_c - Z_t) \quad (5)$$

where, $X_c, Y_c, Z_c = X, Y, Z$ values of the current node; $X_t, Y_t, Z_t = X, Y, Z$ values of the target node

Fruit fly optimization algorithm (FOA) is a new method for finding global optimization based on the food finding behaviour of the fruit fly using their smell and vision capabilities. The algorithm starts with position the fruit flies randomly and finding the best smell concentration (Smell_c), which is given as per distance (Dist_c) of demand point from the current location as given in Eq. (6.1) and (6.2), among the different locations and using the same location for next iteration and calculation of the movement cost. The best smell concentration represents least cost of pipe route. In general, food location is not known, but in this layout optimization, we know the target location, and thus when the target location is reached the smell concentration will be highest and the search will be stopped.

$$\text{Dist}_c = \text{abs}(X_c - X_t) + \text{abs}(Y_c - Y_t) + \text{abs}(Z_c - Z_t) \quad (6.1)$$

$$\text{Smell}_c = \text{Fitness Function} \left(\frac{1}{\text{Dist}_c} \right) \quad (6.2)$$

Dijkstra algorithm is an exhaustive search algorithm to find optimum route. The implementation is same as A* algorithm only the heuristic function, considering the current node to target node, is eliminated. Path finding begin with the start node and moves to the other walkable neighbour nodes (six possible orthogonal movements) next to it until the end node is reached.

Finding an optimal individual pipe route based on pipe route optimization algorithm for a single system does not guarantee optimal multiple pipe system layout, when the 3D space consists of numerous pipe systems. Thus, a metaheuristic algorithm namely simulated annealing is combined with route optimization algorithm to obtain optimal clash-free layout of multiple pipe system in terms of minimum cost.

Global Layout Optimization is done using simulated annealing (SA). SA is a probabilistic technique for approximating the global optimum of a given objective function. It is basically composed of two stochastic processes: firstly, the generation of solutions and secondly the acceptance solutions. SA basically performs by generating a random solution and calculating the cost using the defined objective functions. Followed by generating a new neighboring solution and calculating new cost of the solution using the same objective function for all the multiple pipe system in the layout space as given by Eq. (4). The two results are then compared for the better result, if the better solution is found then it is accepted and marked as current solution otherwise the acceptance of the solution is done by probability function determined by Boltzmann distribution of the cost difference as calculated by Eq. (7). The acceptance criteria of the worse solution depend on the difference between the two solutions and the temperature parameter, as the worse solution are more likely to get accepted at high temperature. The process continues until the termination criteria is met i.e. final temperature is reached.

$$P = e^{\frac{(S_{old} - S_{new})}{K_b T}} \quad (7)$$

where, P = probability function of the acceptance criteria, S_{old} = initial (old) cost solution, S_{new} = new neighboring cost solution, K_b = Boltzmann constant, T = current temperature.

4 Illustrative Example

To illustrate the proposed approach, a typical BIM model of water plant room (see Fig. 4) is studied. Assuming that there is finite number of fixed obstacles and interferences such as architectural and structural elements, equipments, etc. in a 3D layout space. The purpose of multiple pipe layout planning is to obtain clash-free and optimized route for pipe system in the confined space. Dynamo was used to capture the 3D geometric, such as information related to plant room layout space, building components location and dimension as shown in Table 1, and semantic information, like location of supply and demand points, operational requirements and constraints information (branch pipe systems, buffer requirements, accessible clearance, etc.) from BIM model. Dynamo is a visual programming tool, which works as plug-in in Revit [48]. A total of nine routes were used for multiple pipe layout illustration. Supply and demand points of respective routes are shown in Table 2. All the information extracted from Dynamo is stored in excel file, which is further used for pipe layout optimization and determination. Optimization of pipe route depends on a minimum of L_i , NB_j , d_i variables

see Eq. (4), resulting in more than a million possible solutions. Checking each solution will be NP-Hard and unserviceable, as it will increase time and space complexity. Hence, a two heuristic technique namely, 3D A* and FOA, and Dijkstra algorithm that is efficient and reliable to find optimal pipe route is combined with simulated annealing, due to its flexibility and its ability to approach global optimal solution of multiple pipe system layout within a reasonable amount of time. In this example, the developed 3D A*, FOA, Dijkstra - SA approach-based algorithm was implemented in Visual Basic Application in Excel.

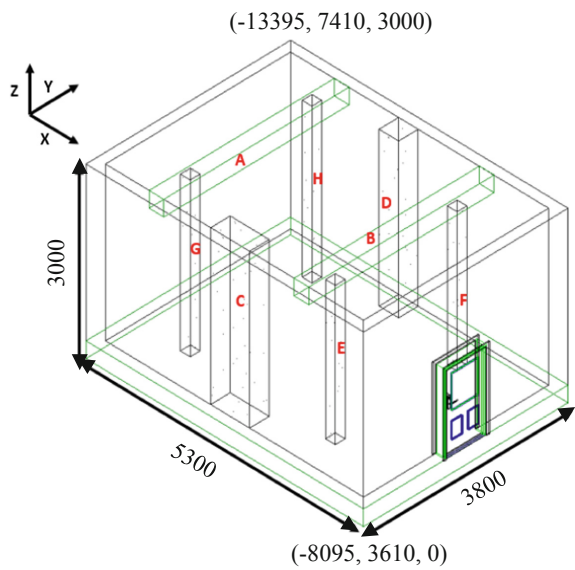


Fig. 4. Typical example of water plant room

Table 1. 3D geometric information of the layout space

S. No.	Object	(X, Y, Z) minimum coordinate value (mm)	(X, Y, Z) maximum coordinate value (mm)
1)	Plant room	(-13395, 3610, 0)	(-8095, 7410, 3000)
2)	Beam A	(-12497, 3610, 2700)	(-12197, 7410, 3000)
3)	Beam B	(-9520, 3610, 2700)	(-9220, 7410, 3000)
4)	Column C	(-11175, 7010, 0)	(-10775, 7410, 3000)
5)	Column D	(-11235, 3610, 0)	(-10435, 4010, 3000)
6)	Column E	(-9474, 6710, 0)	(-9274, 6910, 3000)
7)	Column F	(-9474, 4210, 0)	(-9274, 4410, 3000)
8)	Column G	(-12454, 6710, 0)	(-12254, 6910, 3000)
9)	Column H	(-12454, 4210, 0)	(-12254, 4410, 3000)

Table 2. Supply and demand points for pipe systems

Route. No.	Supply point (start)	Demand point (end)
R1.	(-9000, 3800, 2000)	(-13000, 3800, 2000)
R2.	(-9000, 3800, 2000)	(-13000, 3800, 1000)
R3.	(-13300, 4800, 2000)	(-8800, 3800, 2500)
R4.	(-8500, 7200, 1600)	(-9000, 3700, 2600)
R5.	(-9000, 3700, 2800)	(-10000, 7200, 800)
R6.	(-12000, 7200, 500)	(-11500, 6700, 2500)
R7.	(-10300, 7200, 2500)	(-13300, 5200, 500)
R8.	(-9800, 7200, 1500)	(-13300, 6700, 500)
R9.	(-9500, 7000, 500)	(-9100, 6600, 2500)

A flow chart depicting the developed approach for the multiple routing layout is given in Fig. 5. To determine the pipe route, the room layout space was converted to graph of $100 \text{ mm} \times 100 \text{ mm} \times 100 \text{ mm}$ grid size. A random order of all the pipe routes was selected and used for multiple pipe layout generation. The pipe routing of each system starts in the order of the selected order based on the adopted pipe route optimization algorithm. The path finding starts from the start node, to find the next clash-free optimal node for the path, total six orthogonal neighbor nodes were checked, and their respective total movement cost as mentioned in Eq. (4). was calculated. The neighbor node with the lowest movement cost was chosen to be next potential search node. The search for the next potential node continues until the target node was reached. The nodes that obstruct the building objects or already exists in some other pipe routes are discarded and marked as unwalkable. A new strategy is proposed to avoid the clashes among different pipe routes. The nodes that clash with already existing pipe routes are restricted and discarded for the new pipe routes. Any node that clashes with the start and end points (nodes) of the other pipe routes are marked as unwalkable except if the multiple pipe routes have same start points or end points.

After finding all the pipe routes as per respective selected order. The total movement cost for the entire multiple pipe system layout is calculated and is marked as the current cost. Following which new random order of pipe route is generated and is selected to determine new pipe routes and the new multiple pipe system layout cost. Subsequently, metropolis cycle and the temperature cycle are also updated. The process continues until the stopping criteria (final temperature) is reached and resulting in the multiple pipe layout having minimum layout cost. The simulated annealing parameters used, and their respective values are presented in Table 3.

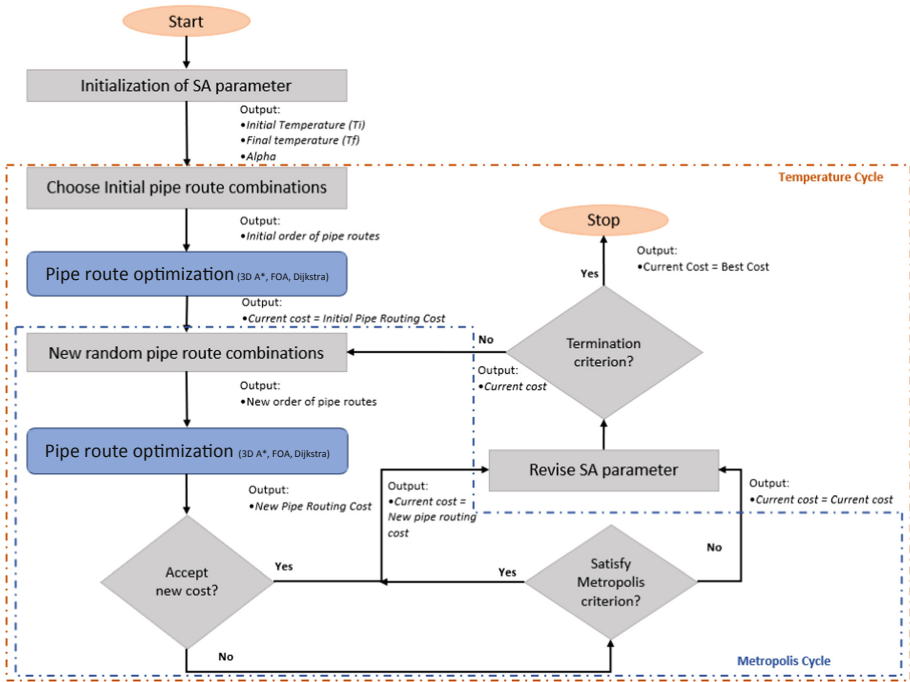


Fig. 5. The developed approach for the multiple routing layout optimization

Table 3. Simulated annealing parameter

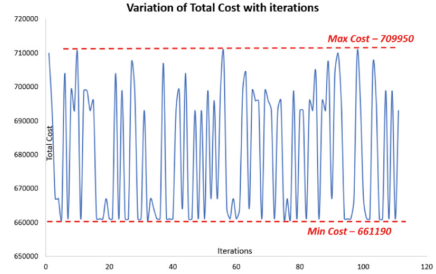
S. No.	Parameter	Value
1.	Initial temperature (T_i)	0.1
2.	Final temperature (T_f)	10
3.	Alpha (α)	0.9
4.	Cooling schedule equation	T_{i+1} (current temperature) = $\alpha * T_i$

The output of the developed approach using all the above-mentioned path route optimization algorithms (3D A*, FOA, Dijkstra) - SA approach for multiple pipe system layout are compared (Table 4) among each other based on cost of pipe layout and time taken by the algorithm. The variation of cost curve and convergence curve of all the algorithms with each iteration is shown in Fig. 6 and Fig. 7. While implementation of 3D A* algorithm for pipe route optimization, four possibilities were considered to know the how cost is comprised with time, as represented in Table 4. For this purpose, the heuristic cost (h_C) function used in 3D A* algorithm was multiplied by factor of 10000, 1000, 100, 1 (original 3D A*). Higher the factor value, lesser is the time to run the algorithm. The 3D A* results are also compared with FOA and Dijkstra results as shown in Table 4. After comparing the respective algorithms results it was found that 3D A with $1 * h_C$ shows best result with minimum total layout cost which is

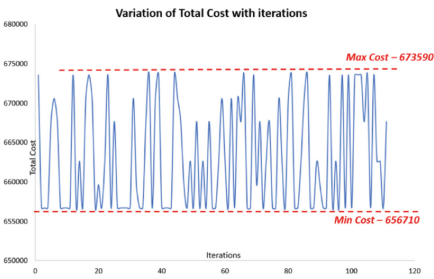
same as Dijkstra algorithm i.e. \$644450 and takes approximately 50% lesser time than the Dijkstra algorithm. The output of the multiple pipe system layout is shown in Fig. 8.



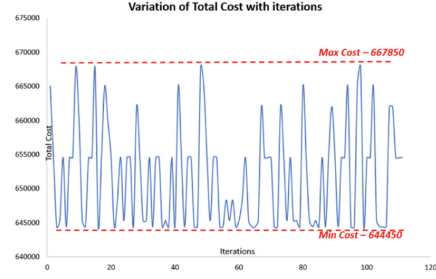
3D A* X 10000 Heuristic cost (hc)



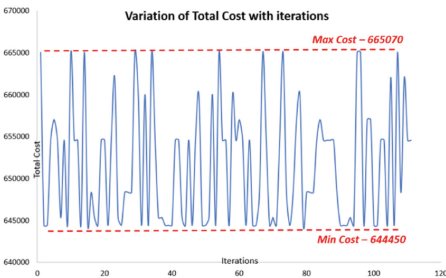
3D A* X 1000 Heuristic cost (hc)



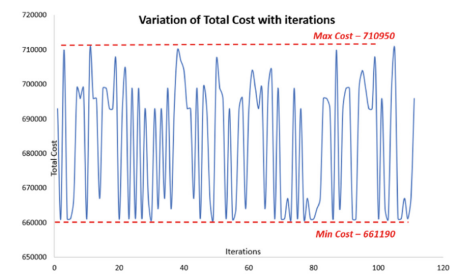
3D A* X 100 Heuristic cost (hc)



3D A* X 1 Heuristic cost (hc)



Dijkstra Algorithm



FOA Algorithm

Fig. 6. Comparison of variation cost of multiple pipe system layout

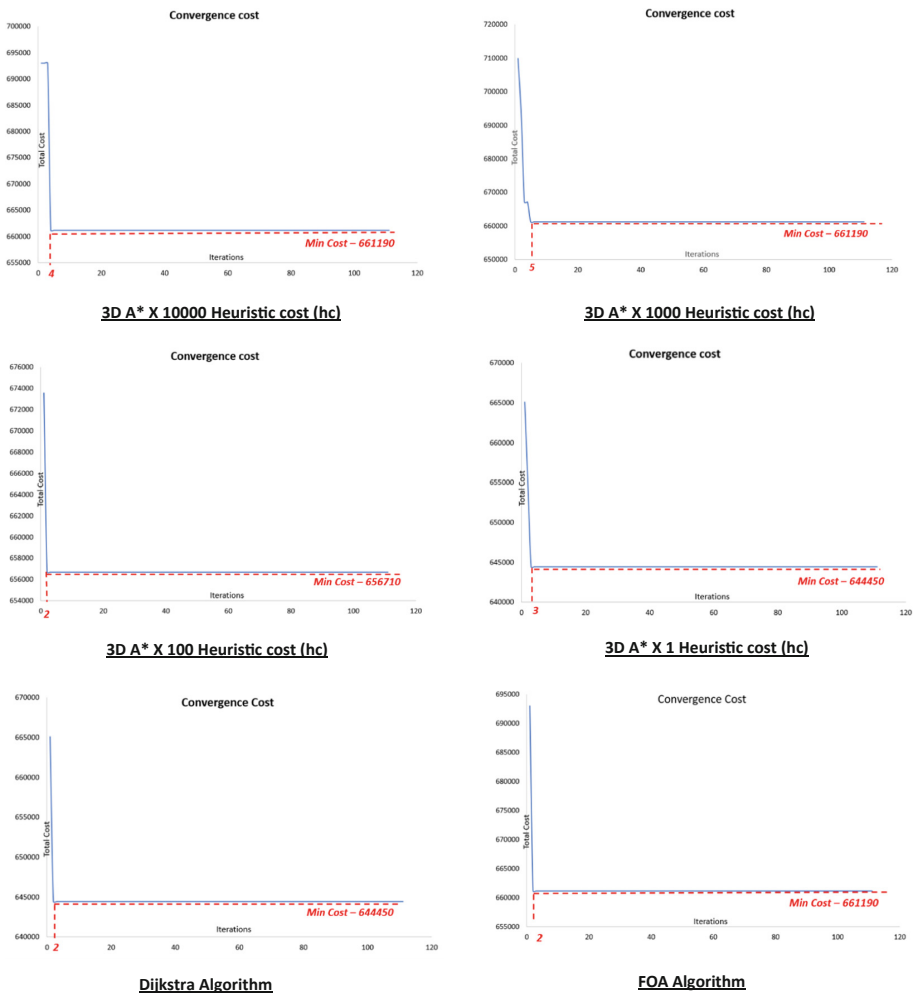


Fig. 7. Comparison of convergence cost of multiple pipe system layout

Table 4. Comparison of output of multiple pipe system layout

S. No.	Pipe route optimization algorithm - SA		Cost (\$US)	Time (hh:mm:ss)
1.	3D A* - SA	10000 × heuristic cost	661190	00:37:18
2.		1000 × heuristic cost	661190	00:41:49
3.		100 × heuristic cost	656710	06:07:17
4.		1 × heuristic cost	644450	13:29:31
5.	Dijkstra		644450	28:07:27
6.	FOA		661190	00:33:20

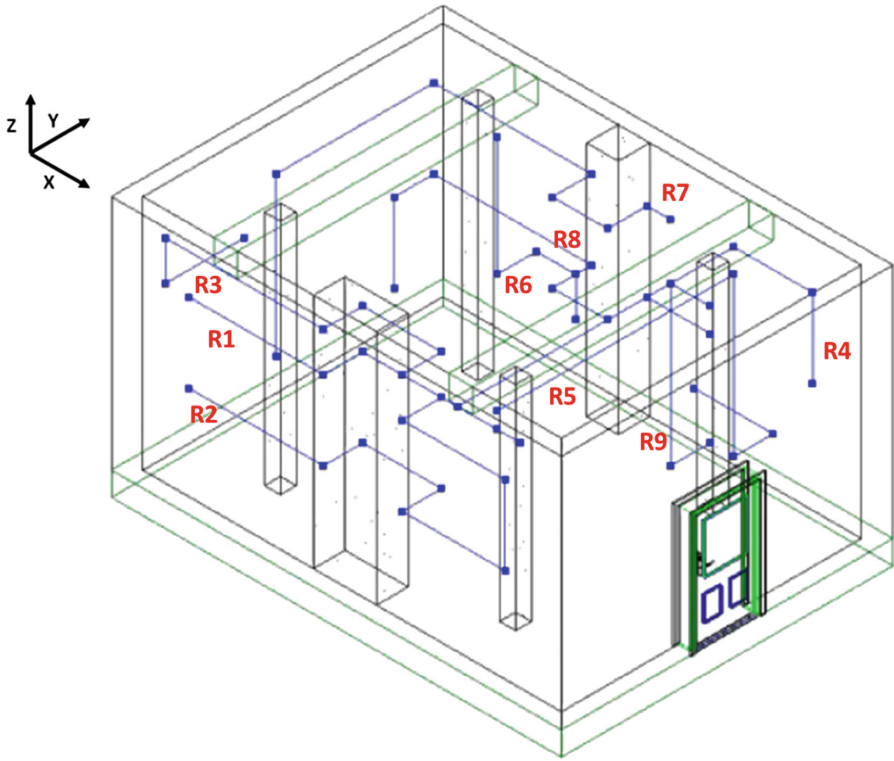


Fig. 8. Output of multiple pipe system layout

5 Conclusion and Future Work

Multiple pipe system layout design and spatial coordination plays an important and challenging role in construction industry. In this paper, a BIM-based approach is proposed to automate multiple pipe system layout optimization in 3D environment using combination of two algorithms. The approach compares pipe route optimization algorithms, Dijkstra algorithm and two heuristic methods (A* and FOA algorithm). The algorithms are modified and used in a directed weighted graph to obtain the multiple pipe system layout. Clashes among pipes and with building components are considered and avoided in the layout optimization. The approach also considers branch pipe routing connecting one supply to many demands points and many supplies to one demand point. The framework is also tested by an illustrative example to further explain the functionalities of the different modules in the framework. After comparing the respective algorithms results it was found that 3D A* shows best result with minimum total layout cost which is same as Dijkstra algorithm and takes approximately 50% lesser time than the Dijkstra algorithm. With the adoption of this framework, time and human effort for the optimized pipe systems design are significantly reduced. The developed framework can also be used in designing of other trades such as mechanical

etc. with further modifications. Therefore, extending the framework for other trades will be considered in the future work.

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